

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	Increase pd until microammeter reads zero. Or decrease pd until microammeter just starts to read a current (1) Reading of voltmeter <u>gives the energy value needed</u> or equivalent statement (1) Accept $E_k = eV_{\text{stop}}$	2			2		2
		(ii)	$\phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{405 \times 10^{-9}} \text{ [J] (1) } - 0.480 \times 1.60 \times 10^{-19} \text{ [J] (1)}$ $\text{or } \phi = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{405 \times 10^{-9} \times 1.60 \times 10^{-19}} \text{ [eV] (1) } - 0.480 \text{ [eV] (1)}$ [3.07] even if not transposed $4.1 \times 10^{-19} \text{ [J] or } 2.6 \text{ [eV] (1)}$		3		3	3	3
		(iii)	For 650 nm, photon energy = $3.1 \times 10^{-19} \text{ J or } 1.9 \text{ eV (1)}$ Photon energy not enough to knock out electron or photon energy < ϕ (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency Alternative Calculation of $f_0 = 6.2 \times 10^{14} \text{ [Hz] ecf}$ or calculation of $\lambda_0 = 4.8 \times 10^{-7} \text{ [m] ecf (1)}$ $4.6 \times 10^{14} \text{ [Hz]} < 6.2 \times 10^{14} \text{ [Hz] or } 650 \text{ nm is } > 480 \text{ nm}$ so no emission occurs (1) More photons [per second] won't help or increasing intensity doesn't increase photon energy (1) Accept only gives more photons at the same energy or doesn't change frequency - Subtract 1 mark if no useful reference made to photons			3	3		3

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	(b)	(i)	$v = \frac{h}{m\lambda}$ transposed and used (1) $v = 1.5 \times 10^7 \text{ [m s}^{-1}\text{]} (1)$		2		2	2	
		(ii)	Electrons are wavelike or diffract or interfere (1) Accept reference to wave-particle duality Graphite crystals act as diffraction gratings or electrons diffract <u>between (around) atom[s]</u> (through lattice) or λ comparable with gaps between atoms (1)	2			2		
			Question 7 total	4	5	3	12	5	8